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What the Yerkes-Dodson Law Says About Stress and Performance

Medically reviewed by Marney A. White, PhD, MS, Psychology — Written by Ann Pietrangelo on October 22, 2020

Stress-performance bell curve | How the law works | Factors |
Optimal arousal or anxiety | Yerkes-Dodson law and the pandemic | Summary

What is the Yerkes-Dodson law?

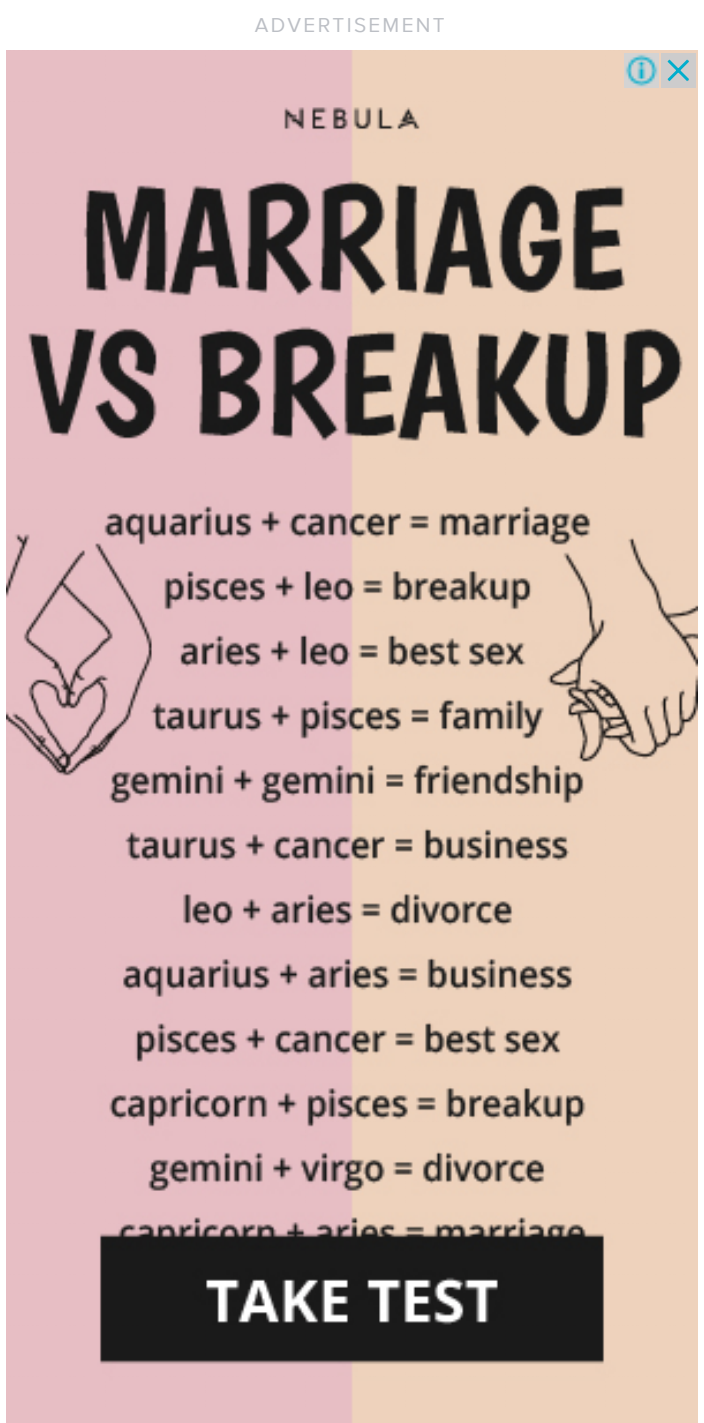
The Yerkes-Dodson law is a model of the relationship between stress and task performance.

It proposes that you reach your peak level of performance with an intermediate level of stress, or arousal. Too little or too much arousal results in poorer performance.

This is also known as the inverted-U model of arousal.

The theory has been around since 1908, when psychologists Robert Yerkes and John Dillingham Dodson performed experiments on mice. Even though Yerkes-Dodson is called a law, it's not a scientific law as much as a psychological concept.

Let's delve a little deeper into how the Yerkes-Dodson law relates to stress and anxiety, as well as its relevance during the COVID-19 pandemic.



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Yerkes-Dodson curve

The Yerkes-Dodson law can be depicted as an upside-down U-shaped curve.

The left side of the curve represents low arousal, or stress. The right side represents high arousal. And at the center is a medium level of arousal.

The vertical line on the left side goes from poor performance (at the bottom) to peak performance (at the top).

The optimal state of arousal and optimal performance come together in the middle of the curve.

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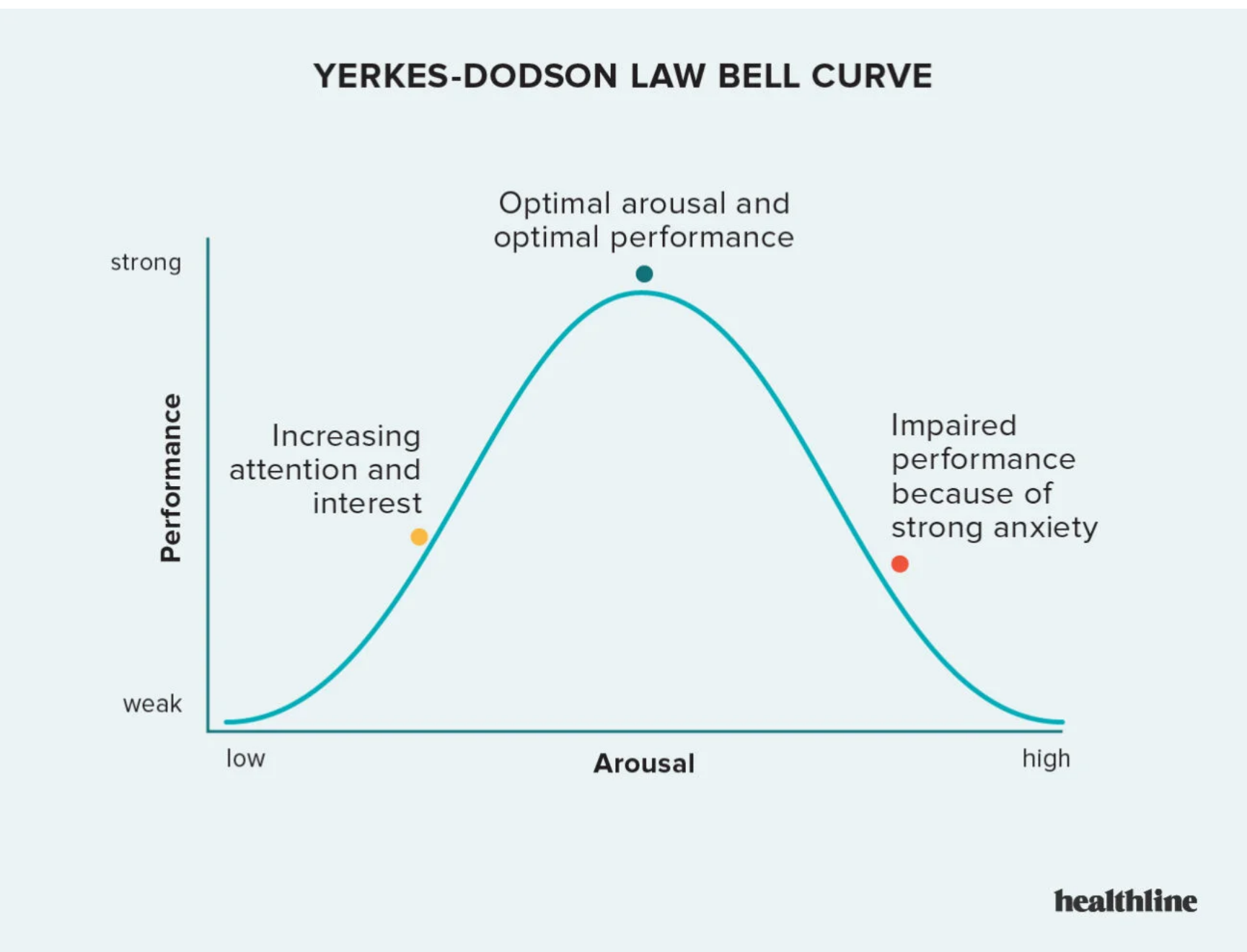

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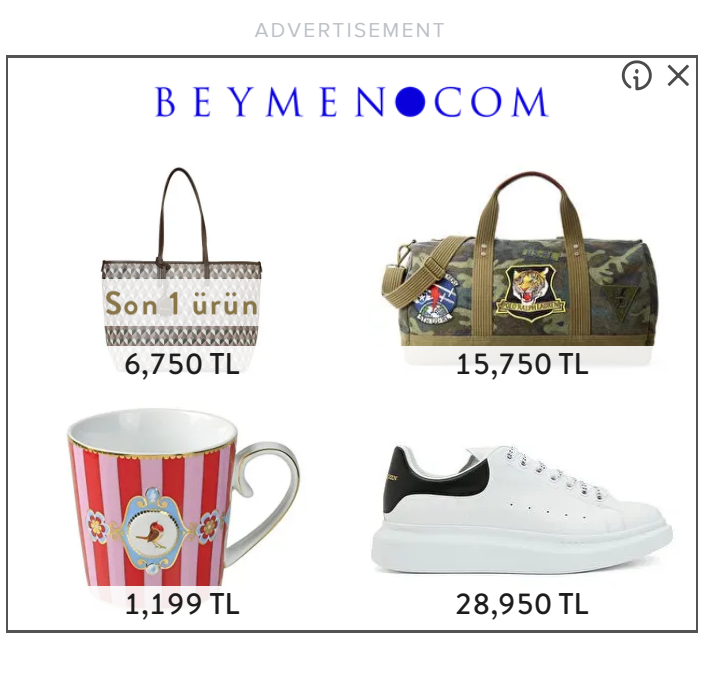

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Yerkes-Dodson law psychology

In discussing the Yerkes-Dodson law, the word "arousal" relates to stress and motivation.

Yerkes and Dodson formed their theory while performing experiments on mice. Using mild electric shocks, they were able to teach the mice to learn a task, which they called a "habit," more quickly.

But as the shocks got stronger, the mice took longer to learn the task, maybe because they were more focused on avoiding the shock than on completing the task.

Yerkes and Dodson hypothesized that as arousal increases, the ability to form a habit or perform a task well also increases. It gives you sufficient motivation.

But that works only up to a certain point, known as the optimal level. As arousal surpasses that point, your performance begins to deteriorate. You're too stressed and anxious to do your best.

So, what's the optimal amount of arousal? That depends on the task. A simpler task requires a higher amount of arousal, while a more challenging task requires a lower level of arousal.

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Low arousal

Having no stress at all isn't necessarily a good thing in terms of performance.

For example, when your job is all about routine and nothing ever changes, boredom sets in. There's no stress, but there's also no motivation. You're not being challenged and have no incentive to go above and beyond. Your work feels meaningless, so you do the bare minimum.

Think about mice in a maze with no electric shocks and no cheese at the end. There's no reason to put in the effort to navigate the maze.

Optimal arousal

A moderate level of stress goes a long way. It's manageable, motivational, and performance enhancing. Your heart beats a bit faster. You feel a sense of clarity and alertness. Your brain and body are all fired up.

It's that little extra push you need when a hard deadline looms and you're up for a promotion. It's the rush you get before the black belt test you've been working toward for so long.

There's something you want. You've definitely got skin in the game. And a moderate surge of stress is boosting your performance.

High arousal

Intense stress can lead to a fight, flight, or freeze response.

It's the final play of the season, winner take all, and you're up to bat. It's the make-or-break project that could get you a life changing bonus. It's the test that could keep you from graduating.

In these types of situations, stress and anxiety are ramping up to an unmanageable level.

Your heart may beat faster, but it's unsettling, distracting, even nerve-wracking. You've lost focus and you're not reaching your full potential.

You're all too aware that you've got skin in the game, but it's working against you. It's too much.

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Factors that affect the level of performance under pressure

We all experience stress differently, so the optimal level of stress for you won't be the same as it is for someone else. Many factors are at play, including:

Skill level

If something is new to you, that alone may be enough to provide a challenge. You're interested and have a few things to learn, so you're naturally motivated.

If you've been doing something for a while and have gained a higher skill level, it's easy to feel bored. You may need a little incentive to perk up again.

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Personality

What's just the right amount of stress for one person may be an overwhelming level of stress for another. Some people perform better under pressure than others.

Depending on the task at hand, it may matter if you're naturally introverted or extroverted. Your life experiences, beliefs, and fears may all play a role in how susceptible you are to stress and how you deal with stress.

Confidence

You may perform better under pressure if you have a lot of self-confidence to begin with. You're filled with positive thoughts and an "I've got this" mindset.

If you lack self-confidence, have negative thoughts, and tend to self-criticize, you may not be able to perform your best when the pressure is on.

Task complexity

You can probably perform simple tasks very well, even when you're under a lot of pressure. But you'll probably benefit from a low-pressure environment when tackling a complex or unfamiliar task.

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What levels of stress are optimal for best performance?

Basically, you need just enough stress to provide motivation but not so much that you're overwhelmed. That's difficult to pinpoint and will differ from person to person.

A mild to moderate amount of short-term stress can result in an acute stress response that provides the motivation and energy you need, just when you need it. It lasts only long enough to help you perform your best.

Chronic stress is another matter entirely and isn't likely to benefit you at all. In fact, a number of the effects of stress can negatively impact your physical and mental health.

The inverted-U curve looks a little different for each person and probably even changes at different points in your life.

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Yes No

How the Yerkes-Dodson law relates to the COVID-19 pandemic

Quarantine fatigue. Pandemic fatigue. Whatever you call it, it's a very real phenomenon: Restricted social activities, boredom, and lack of structure can rob you of motivation.

On the other hand, you may be feeling overly stressed and anxious about:

- your and your family's health
- lack of work and associated financial problems
- the coming together of work, school, and living spaces
- unpredictability and managing expectations for the future
- lack of physical activity
- overstimulation from the news
- stress cues from others

While we work our way through the pandemic, you may feel both understimulated and overstimulated. Charting your own inverted-U curve during this trying time may be difficult.

Takeaway

The Yerkes-Dodson law is the theory that there's an optimal level of arousal that results in optimal performance.

Too little arousal doesn't provide much in the way of motivation. Too much arousal causes a stronger stress reaction that can hamper performance.

That optimal level of arousal differs from person to person, according to factors like the specific task, degree of skill, and confidence level.

Getting to that optimal arousal zone can be difficult because some factors aren't within your control. But the Yerkes-Dodson law shows that there may be a sweet spot for achieving your best performance.

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